



Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 27 Decision Mathematics 1

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General Comments

The paper was accessible to almost all learners, and many responses demonstrated sound knowledge across the full range of topics.

Many responses were well presented and made effective use of the diagrams and tables provided in the answer booklet.

Learners should be reminded to show their methods clearly, as Decision Mathematics is a method-based examination.

Comments on Individual Questions

Question 1

Part (i)

This was a standard application of Dijkstra's algorithm and many learners scored full marks, getting the correct quickest path and shortest time. Some responses did not show sufficient attention to the order of working values. Learners should be reminded to show their working values clearly and not to cross them out when they are replaced by smaller values.

If mistakes were made, this was most common with the working values at E (62 calculated incorrectly), G (68 calculated incorrectly) or J (missing values). Other marks were sometimes not awarded where there were errors in the order of labelling, particularly at BC and GI, or where values were missing. The path and length were generally correct, even when there were errors on the diagram.

Part (ii)

Some responses gave the correct answer in seconds, while others did not use the order of n^2 correctly in their calculations. A common error was to leave the answer as 16200 seconds, rather than converting it into minutes as required by the question. Learners should be reminded to check the required units carefully.

Question 2

Part (i)(a)

Most learners used Kruskal's algorithm to find the minimum spanning tree, with very few incorrectly using Prim's algorithm or even the nearest neighbour algorithm. Learners used a variety of methods to document their working, with the list of arcs in order with ticks/crosses being the most efficient. A few learners used a bubble sort to put their arcs in order, wasting valuable time. Many learners wrote out the full list of arcs in order and then wrote the list of selected arcs, not explicitly showing their rejections and did not gain the final CSO mark. A few learners were not successful in finding GH as the second smallest arc and did not gain any marks. Some learners drew the tree. While not required by the question, this can be helpful to avoid creating a cycle when adding arcs.

Part (i)(b)

Those learners finding the MST successfully in part a), added up the weights accurately with few numerical errors seen.

Part (ii)(a)

Generally, learners used the correct four nodes for the pairings, although some learners attempted to use BFGH, CFGH, BCEH or DHCF rather than the correct set of BCFH. Those learners with the correct pairings generally found at least one total correctly, although some learners did not calculate the total for all (or any) pairings and marks were not awarded. Common errors saw the total for BF and CH calculated as 105 (instead of 104) and the total for BH and CF calculated as 70 (instead of 69). Some responses did not write out the list of arcs in full, and the final accuracy mark was not awarded.

Part (ii)(b)

Learners who had correctly paired the nodes and calculated the three totals, generally calculated the length accurately from their shortest pairing.

Question 3

Part (a)

This part was generally answered well, with most learners scoring at least two out of the three available marks. Very few ascending sorts or quick sorts were seen, and most learners were able to show just the final results of each pass rather than every comparison. A small number of learners chose to write out every swap, and not just the result at the end of the pass, which was very time consuming for them. The most common error was to just stop after the 3rd pass, when the numbers were in order, with no 4th pass shown; this led to the final mark not be awarded. Some responses used a “cross arrow” notation to show all the swaps in a single line. This method was sometimes difficult to follow and could lead to errors in one or more passes.

Part (b)

This part of the question was more challenging. Most learners managed to find the bin totals and deduce that n was 58 or above, although some excluded 58. There were several possible ways to form an inequality leading to $n < 61$. However, some responses gave higher upper bounds for n . Some learners identified the correct inequalities but did not show sufficient working, which limited access to the marks.

Question 4

Part (a)

This was generally done well with most learners obtaining at least the first B mark for the immediately preceding activities for H, I and J. Errors tended to be in the immediately preceding activities for L where either J or G was regularly omitted.

Part (b)

This part was accessible for most learners, and many correct diagrams were seen. Errors most often occurred with the late event time, particularly at the end of C, where 9 was sometimes given instead of 4.

Some responses did not use the dummy from the end of C to the end of B to obtain early and late event times of 4 for the end of B. The dummies from the end of G to the end of F, and from the end of F to the end of J, also caused difficulty in some responses. Where these dummies were not processed correctly, several subsequent values were affected, and the three marks were not accessed.

Part (c)

Learners frequently included B instead of C in the critical activities as they did not take into account the length of B and C.

Part (d)

Most learners drew a Gantt (cascade) chart, although a few schedules and some blank entries were seen. The critical activities were generally drawn at the top of the diagram, although errors tended to occur with the B/C critical activity. Non-critical activity F caused issues, where many learners with the correct diagram only gave a float of 2 (to 16 the early event time). K also caused some issues, with learners starting K at 15 or 20 instead of 16 and a float ending too soon. A few learners omitted an activity from the diagram. Those learners who made errors in their late times in (b) used these times for the floats, and the marks were not accessed. Learners should ensure that they draw the floats clearly, ideally by shading, and not just with a faint dashed line.

Question 5

Many learners made an attempt at this question, although there were a few blank scripts seen. Most learners found the objective function in x , y , and z and those that substituted for z , generally found the correct objective function; some responses did not state 'minimise' for their new objective. Having a fixed total for the number of cups caused some issues, with learners expecting an inequality and did not translate the description in an equation. This meant they did not have an expression for z and either used $z = 3/5 y$ or rearranged the objective function which they had equated to 200.

The constraint to buy at most 5 medium cups for every 3 large cups was frequently seen with $5y$ and $3z$ instead of $3y$ and $5z$ and some learners had the inequality the wrong way around, meaning that they could not access the accuracy mark for their substitution of z .

Most learners successfully translated the number of large cups being at most 35% of the total of small and medium cups, but some learners used x , y and z instead of just x and y . Some learners had the inequality the wrong way around, meaning that they could not access the accuracy mark for their substitution of z .

Some of the working seen was confused, with a variety of constraints and objective functions not clearly laid out. The best responses produced a clear statement of the final objective and the two simplified constraints.